

IMAT MOCK EXAM 8 — "Final Rehearsal"

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 44/90.0 · Mock exam

Section A - Logical Reasoning & Reading (Q1-9)

1. "The human gut hosts trillions of microbes that assist digestion, synthesise vitamin K and several B vitamins, and help train the developing immune system. A course of antibiotics can transiently reduce the diversity of this community. In mice, transplanting gut microbes from obese donors transfers some of the donors' metabolic traits to lean recipients, and diet can reshape the community's composition within days."

Which of the following is not supported by the passage?

- A)** Gut microbes influence their host's metabolism.
- B)** Diet can change the composition of the gut community within days.
- C)** Gut microbes synthesise some vitamins.
- D)** Taking probiotic supplements cures inflammatory bowel disease.
- E)** Antibiotics can temporarily reduce gut microbial diversity.

2. A city finds that neighbourhoods with more street trees have lower childhood asthma rates. The council concludes that planting trees will reduce asthma.

Which of the following, if true, most weakens the council's conclusion?

- A)** Tree-lined streets are concentrated in wealthier neighbourhoods, which also have better housing and healthcare.
- B)** Asthma has many triggers.
- C)** Trees produce pollen.
- D)** Trees are planted on wide streets.
- E)** Some neighbourhoods have parks rather than street trees.

3. (General knowledge) The Milano Cortina 2026 Winter Olympic Games are the __ time that Italy has hosted the Winter Olympic Games.

- A)** fourth **B)** first **C)** second **D)** fifth **E)** third

4. A sealed box contains a number of levers. Exactly one of the following two statements is true:

- (1) All of the levers in the box are made of steel.
- (2) None of the levers in the box is made of steel.

Which conclusion follows necessarily?

- A)** No lever is made of steel.
- B)** The box contains at least one lever.
- C)** All the levers are made of steel.
- D)** Some, but not all, of the levers are made of steel.
- E)** The box is empty.

5. Three tasks (triage, laboratory and records) must be assigned, one each, to Anil, Bea and Cara. Anil refuses to do triage. Cara refuses to do records.

How many different valid assignments are possible?

- A)** 2 **B)** 3 **C)** 4 **D)** 5 **E)** 6

6. A clock reads 7:35. What is the angle between the hour hand and the minute hand?

- A)** 12.5° **B)** 17.5° **C)** 20° **D)** 22.5° **E)** 35°

7. In a dose-finding trial, patients rated their average pain on a 0 to 10 scale. At doses of 0, 2.5, 5 and 10 mg the mean pain scores were 8, 5, 3 and 3 respectively.

Which conclusion is best supported?

- A) Higher doses are always better.
- B) The drug is unsafe at 10 mg.
- C) Increasing the dose beyond 5 mg produces no further average reduction in pain.
- D) The drug has no effect on pain.
- E) 10 mg is the optimal dose.

8. A share price rises by 10% one day and falls by 10% the next day. The net change over the two days is:

- A) a 1% decrease B) no change C) a 1% increase D) a 10% decrease E) a 19% decrease

9. In a class of 30 students, 18 study biology, 15 study chemistry, and 8 study both. How many study neither subject?

- A) 2 B) 3 C) 4 D) 5 E) 6

Section B - Biology (Q10-32)

10. A father is blood group AB; a mother is blood group O (genotype OO). Which statement about their children is correct?

- A) All children are AB.
- B) The four ABO phenotypes are all equally possible.
- C) All children are O.
- D) The children's groups cannot be predicted.
- E) Each child must be A or B (genotypes AO or BO); AB and O are impossible.

11. Genes A and B lie on the same chromosome and crossing over between them never occurs. An individual with chromosomes AB/ab (coupling arrangement) is crossed with ab/ab. The offspring will be:

- A) only the two recombinant types
- B) offspring in a 9:3:3:1 ratio
- C) only the two parental types, AB/ab and ab/ab, in equal proportions
- D) all offspring AB/ab
- E) all four phenotypic classes in a 1:1:1:1 ratio

12. A woman is a carrier of haemophilia ($X^H X^h$). Her husband has haemophilia ($X^h Y$). What fraction of their daughters is expected to have haemophilia?

- A) 0 B) 1/4 C) all of them D) 1/2 E) 3/4

13. In double-stranded DNA, guanine pairs with cytosine through:

- A) three hydrogen bonds
- B) one hydrogen bond
- C) four hydrogen bonds
- D) a covalent bond
- E) two hydrogen bonds

14. Retroviruses such as HIV copy their RNA genome into DNA inside the host cell using:

- A) DNA polymerase of the host
- B) restriction enzymes
- C) ribozymes
- D) RNA polymerase
- E) reverse transcriptase

15. During translation, the mRNA codon 5'-AUG-3' is read by the tRNA anticodon. Written in the 3' to 5' direction, the anticodon is:

- A) 3'-UAC-5' B) 3'-AUG-5' C) 3'-TAC-5' D) 5'-UAC-3' E) 5'-ATG-3'

- 16.** Which protein exhibits quaternary structure?
- A)** ribonuclease, a single 124-residue chain
 - B)** pepsin, a single-chain enzyme
 - C)** haemoglobin, with two α and two β chains
 - D)** myoglobin, a single 153-residue chain
 - E)** an α -helix
- 17.** When two amino acids join to form a dipeptide, the reaction is:
- A)** hydrolysis, consuming water
 - B)** a condensation reaction releasing one molecule of water and forming a peptide bond
 - C)** a substitution of the R groups
 - D)** a neutralisation releasing salt
 - E)** an oxidation releasing hydrogen
- 18.** Fermentation yields only about 2 ATP per glucose, versus roughly 30 by full aerobic oxidation, because:
- A)** NAD^+ is never regenerated in fermentation
 - B)** the electron transport chain runs backwards during fermentation
 - C)** glycolysis stops when fermentation begins
 - D)** fermentation produces no ATP at all
 - E)** the glucose is only partially oxidised: most of its energy remains trapped in lactate or ethanol
- 19.** Substrate-level phosphorylation (direct transfer of a phosphate from a substrate to ADP) occurs in:
- A)** pyruvate oxidation only
 - B)** the electron transport chain only
 - C)** both glycolysis and the Krebs cycle
 - D)** glycolysis only
 - E)** the Krebs cycle only
- 20.** Leaves look green because chlorophyll:
- A)** is a green pigment that absorbs all wavelengths equally
 - B)** destroys green light
 - C)** absorbs only green light
 - D)** absorbs red and blue light strongly and reflects/transmits green light
 - E)** emits green fluorescence under sunlight
- 21.** The sodium-potassium pump of the cell membrane moves:
- A)** 3 Na^+ out for every 2 K^+ in, at the cost of one ATP
 - B)** 2 Na^+ out for every 3 K^+ in
 - C)** Na^+ and K^+ both down their gradients
 - D)** 3 K^+ out for every 2 Na^+ in
 - E)** equal numbers of Na^+ and K^+ in opposite directions
- 22.** Which checkpoint ensures that DNA replication is complete and undamaged before mitosis begins?
- A)** the M checkpoint (spindle)
 - B)** the G2 checkpoint
 - C)** the S checkpoint
 - D)** cytokinesis
 - E)** the G1 checkpoint
- 23.** Meiosis II resembles mitosis because in both processes:
- A)** the cell becomes haploid for the first time
 - B)** bivalents align on the equator
 - C)** sister chromatids separate and move to opposite poles
 - D)** homologous chromosomes pair and cross over
 - E)** the chromosome number is halved

- 24.** The inside of a resting neurone is negative relative to the outside mainly because:
- A)** the membrane is impermeable to all ions at rest
 - B)** chloride ions enter the cell continuously
 - C)** large anions are actively transported out
 - D)** potassium ions diffuse out through leak channels down their gradient, leaving unbalanced negative charge inside
 - E)** sodium ions are pumped into the cell
- 25.** Which sequence correctly traces a simple reflex arc?
- A)** receptor → CNS → effector → motor neurone → sensory neurone
 - B)** effector → motor neurone → CNS → sensory neurone → receptor
 - C)** receptor → motor neurone → CNS → sensory neurone → effector
 - D)** CNS → receptor → sensory neurone → motor neurone → effector
 - E)** receptor → sensory neurone → CNS → motor neurone → effector
- 26.** Which hormone raises blood glucose as part of the acute stress response?
- A)** ADH (vasopressin)
 - B)** insulin
 - C)** oxytocin
 - D)** aldosterone
 - E)** adrenaline (epinephrine)
- 27.** Vaccination with a live attenuated (weakened) pathogen produces which type of immunity?
- A)** active artificial immunity, with the person's own plasma cells and memory cells
 - B)** natural passive immunity via the placenta
 - C)** natural active immunity after catching the disease
 - D)** passive artificial immunity from the injected antigens themselves
 - E)** no immunity, since the pathogen is weakened
- 28.** The natural pacemaker region that initiates each heartbeat is the:
- A)** bundle of His
 - B)** vagus nerve
 - C)** sinoatrial (SA) node
 - D)** atrioventricular (AV) node
 - E)** Purkinje fibres
- 29.** Gas exchange in the pulmonary alveoli is highly efficient mainly because:
- A)** the alveoli are muscular and pump air
 - B)** each alveolus contains cartilage rings
 - C)** the alveoli store oxygen like a tank
 - D)** their enormous total surface area is combined with an extremely thin (squamous) wall
 - E)** alveolar air is under high pressure
- 30.** ADH (vasopressin) increases water reabsorption in the kidney by:
- A)** constricting the efferent arteriole
 - B)** inserting aquaporin channels into the collecting duct wall
 - C)** opening sodium channels in the loop of Henle
 - D)** raising glomerular filtration pressure
 - E)** dissolving urea in the filtrate
- 31.** PCR (the polymerase chain reaction) requires a heat-stable DNA polymerase because:
- A)** the enzyme must work at refrigerator temperature
 - B)** DNA cannot be copied at low temperatures
 - C)** the primers are destroyed by heat and must be resynthesised
 - D)** heat is needed to insert the product into bacteria
 - E)** each cycle includes a high-temperature step (about 95 °C) that separates the DNA strands

32. An antibiotic blocks the bacterial 70S ribosome but not the eukaryotic 80S ribosome. This selectivity means the drug can:
- A) work only against viruses
 - B) act on the patient's mitochondria only
 - C) harm bacteria without directly stopping protein synthesis in the patient's own cells
 - D) be given at any dose without side effects
 - E) kill all cells equally

Section C - Chemistry (Q33-47)

33. What is the oxidation number of carbon in methanol, CH_3OH ?
A) -4 B) -2 C) 0 D) +2 E) +4
34. In the reaction $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaOCl} + \text{H}_2\text{O}$, the chlorine:
A) is simultaneously oxidised and reduced (disproportionation): $0 \rightarrow -1$ in NaCl and $0 \rightarrow +1$ in NaOCl
B) is reduced only
C) is oxidised only
D) keeps oxidation number 0 in both products
E) is unchanged, only dissolved
35. Which molecule has zero net dipole moment despite having polar bonds?
A) NH_3 B) SO_2 C) H_2O D) BF_3 E) CHCl_3
36. Which molecule has a trigonal pyramidal shape?
A) H_2O B) NH_3 C) BF_3 D) CO_2 E) CH_4
37. What is the pH of a 0.005 M solution of $\text{Ba}(\text{OH})_2$ at 25 °C? ($\text{Ba}(\text{OH})_2$ releases 2 OH^- per formula unit; $K_w = 1.0 \times 10^{-14}$)
A) 1.7 B) 2 C) 11 D) 12 E) 13
38. Solution A has pH 4 and solution B has pH 6. The $[\text{H}^+]$ of A compared with B is:
A) 1.5 times greater
B) 2 times greater
C) 10 times greater
D) 100 times greater
E) 1000 times greater
39. How many atoms (total) are present in 0.25 mol of CO_2 ? ($N_a = 6.02 \times 10^{23} \text{ mol}^{-1}$)
A) 0.75×10^{23} B) 1.5×10^{23} C) 3.0×10^{23} D) 4.5×10^{23} E) 6.0×10^{23}
40. Heating 10.0 g of pure CaCO_3 ($M = 100 \text{ g/mol}$) would release 2.24 L of CO_2 at STP. An experiment on an impure sample of the same mass releases 1.68 L at STP. The percentage of CaCO_3 in the sample is:
A) 25% B) 60% C) 67% D) 75% E) 133%
41. A gas at 27 °C and 1.0 atm is sealed in a rigid vessel and heated to 127 °C. The new pressure is closest to:
A) 0.75 atm B) 1.0 atm C) 1.11 atm D) 1.33 atm E) 4.2 atm
42. A chlorine atom is smaller than a sodium atom mainly because:
A) chlorine has more protons pulling on electrons in the same shell
B) chlorine gains electrons, sodium loses them
C) sodium is a metal and chlorine is a gas
D) chlorine's electrons repel each other more strongly
E) chlorine has more electron shells

- 43.** Methanol reacts with propanoic acid under reflux with concentrated H_2SO_4 . The organic product is:
- A)** propanal
 - B)** methanoic acid
 - C)** propanoic anhydride
 - D)** propan-1-ol
 - E)** methyl propanoate
- 44.** Hydrogen bromide adds to propene ($\text{CH}_3\text{-CH=CH}_2$). The major product is:
- A)** propane
 - B)** 1,2-dibromopropane
 - C)** 2-bromopropane
 - D)** 2-bromopropene
 - E)** 1-bromopropane
- 45.** For the heterogeneous equilibrium $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$, the equilibrium constant is written as:
- A)** $K_c = [\text{CaO}][\text{CO}_2]$
 - B)** $K_c = 1$, since solids are constant
 - C)** $K_c = [\text{CO}_2]$
 - D)** $K_c = 1/[\text{CO}_2]$
 - E)** $K_c = [\text{CO}_2]/([\text{CaO}][\text{CaCO}_3])$
- 46.** In a working galvanic (voltaic) cell:
- A)** anions in the salt bridge migrate toward the cathode compartment
 - B)** electrons flow from cathode to anode through the external wire
 - C)** oxidation occurs at the cathode
 - D)** electrons flow through the salt bridge
 - E)** oxidation occurs at the anode and the electrons released flow through the external wire from anode to cathode
- 47.** Four 0.10 M solutions are prepared: glucose ($\text{C}_6\text{H}_{12}\text{O}_6$), NaCl, CaCl_2 , and sucrose. Which has the lowest freezing point?
- A)** CaCl_2 (three ions per formula unit)
 - B)** sucrose (no ions)
 - C)** NaCl (two ions)
 - D)** all four freeze at the same temperature
 - E)** glucose (no ions)

Section D - Mathematics (Q48-54)

- 48.** Solve the inequality $8 - 5x < -12$.
- A)** $x < -4$ **B)** $x > 4$ **C)** $x < 4$ **D)** $x \geq 4$ **E)** $x > -4$
- 49.** Solve $-3 < 2x + 1 \leq 7$.
- A)** $-4 < x \leq 6$ **B)** $-2 \leq x \leq 3$ **C)** $-2 < x < 3$ **D)** $-2 < x \leq 3$ **E)** $-1 < x \leq 4$
- 50.** If $2^x = 40$, then x satisfies:
- A)** $x = 6$ **B)** $4 < x < 5$ **C)** $6 < x < 7$ **D)** $x = 5$ **E)** $5 < x < 6$
- 51.** One card is drawn at random from a standard 52-card deck. What is the probability it is a heart or a king?
- A)** $1/4$ **B)** $1/13$ **C)** $4/13$ **D)** $13/52$ **E)** $17/52$
- 52.** A straight ramp 8 m long makes a 30° angle with the horizontal. How high is its upper end above the ground? ($\sin 30^\circ = 0.5$, $\cos 30^\circ = 0.87$)
- A)** 2 m **B)** 4 m **C)** 6.9 m **D)** 8 m **E)** 16 m

53. If $f(x) = 2x - 1$, then $f(f(x))$ equals:
A) $2x - 3$ **B)** $3x - 4$ **C)** $4x - 3$ **D)** $4x - 1$ **E)** $4x + 1$
54. The circle $(x - 1)^2 + (y + 2)^2 = 16$ has centre and radius:
A) $(1, -2)$, radius 4
B) $(1, -2)$, radius 16
C) $(-1, 2)$, radius 4
D) $(-1, 2)$, radius 16
E) $(1, 2)$, radius 4

Section E - Physics (Q55-60)

55. A 24 V battery drives a $2\ \Omega$ resistor in series with a parallel pair of $12\ \Omega$ and $6\ \Omega$. What current flows from the battery?
A) 2 A **B)** 3 A **C)** 4 A **D)** 6 A **E)** 12 A
56. A 60 W lamp is designed for a 120 V supply. The resistance of its filament is:
A) $0.5\ \Omega$ **B)** $2\ \Omega$ **C)** $60\ \Omega$ **D)** $240\ \Omega$ **E)** $7200\ \Omega$
57. Water boiling at atmospheric pressure stays at exactly $100\ ^\circ\text{C}$ even though the flame keeps supplying heat. This is because:
A) the supplied energy goes into the latent heat of vaporisation, breaking intermolecular attractions instead of raising temperature
B) boiling only occurs without heating
C) the thermometer is faulty above $90\ ^\circ\text{C}$
D) water has no specific heat at $100\ ^\circ\text{C}$
E) the molecules stop moving at $100\ ^\circ\text{C}$
58. Cars once had rigid metal bumpers; modern cars crumple in a crash. The physics reason a crumple zone reduces injury is that it:
A) eliminates the change of momentum
B) extends the collision time, reducing the average force for the same change of momentum
C) converts the passenger into the system's centre of mass
D) reduces the car's momentum to zero instantly
E) increases the impulse delivered
59. On a distance-time graph, a horizontal segment means the object is:
A) moving at constant non-zero speed
B) decelerating to rest at the end of the segment
C) moving backwards
D) accelerating
E) stationary (zero speed for that interval)
60. A block of density $0.90\ \text{g/cm}^3$ floats in water ($1.00\ \text{g/cm}^3$). What fraction of the block's volume is submerged?
A) 10% **B)** 25% **C)** 50% **D)** 90% **E)** 100%

Answer key

This key refers to the option order printed in this paper. The online app shuffles option order on every attempt, so answer letters inside the app will differ from the ones here.

1	D	2	A	3	E	4	B	5	B
6	B	7	C	8	A	9	D	10	E
11	C	12	D	13	A	14	E	15	A
16	C	17	B	18	E	19	C	20	D
21	A	22	B	23	C	24	D	25	E
26	E	27	A	28	C	29	D	30	B
31	E	32	C	33	B	34	A	35	D
36	B	37	D	38	D	39	D	40	D
41	D	42	A	43	E	44	C	45	C
46	E	47	A	48	B	49	D	50	E
51	C	52	B	53	C	54	A	55	C
56	D	57	A	58	B	59	E	60	D